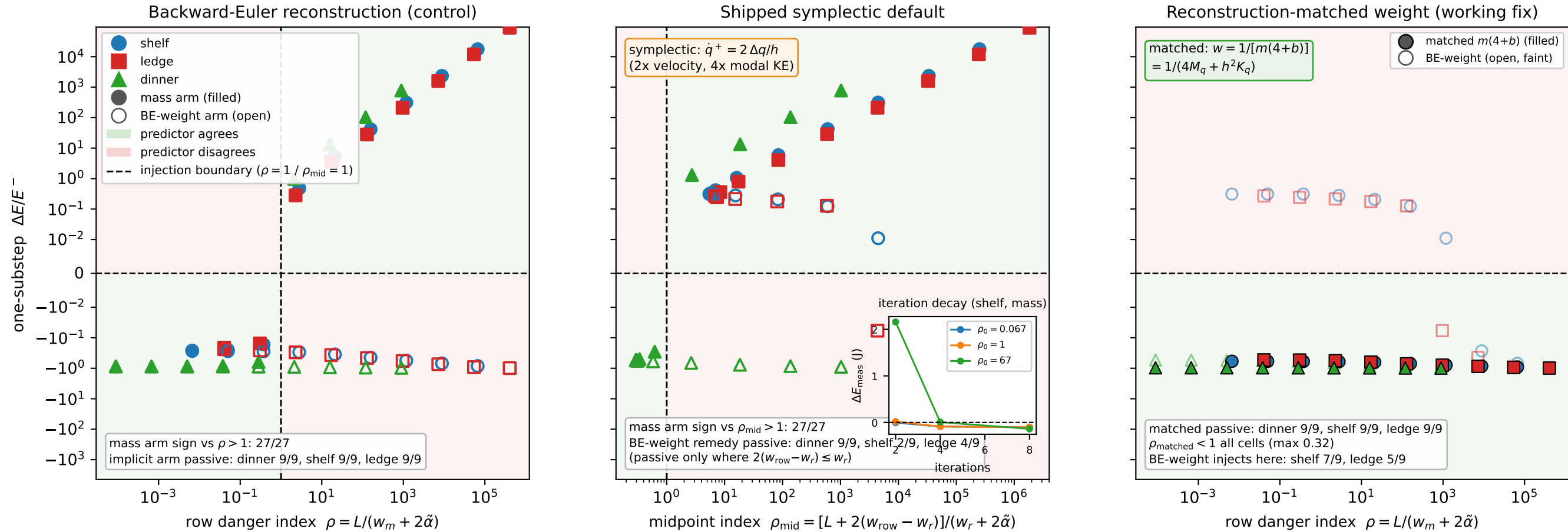


Shipped XPBD support row: each reconstruction against its validated boundary, and the reconstruction-matched fix



Left: the backward-Euler control flips sign at the row danger index $\rho = 1$ (mass arm 27/27; BE-weight remedy passive 9/9 per scene).
 Middle: the shipped symplectic default reconstructs $\dot{q}^+ = 2 \Delta q/h$ and deposits 4x the modal KE, shifting the boundary to the midpoint index $\rho_{\text{mid}} = 1$ (mass arm 27/27); the BE-weight remedy, matched to backward Euler, now injects (passive dinner 9/9, shelf 2/9, ledge 4/9).
 Right: the reconstruction-matched weight $m(4+b) = 4 M_q + h^2 K_q$ restores passivity at every host danger index (27/27; $\rho_{\text{matched}} < 1$), at the same operating points where the BE-weight injects (open, faint). Hard contact. Hypotheses: one sweep, cold start, $e = 0$.